

Redemption Architecture Executive Summary

This document summarizes the architecture, trade-offs, and operational plan for the Redemption microservice.

1. Architecture: EKS cluster in 3 AZs with private API endpoint, ALB ingress, Aurora Multi-AZ, Redis, and S3.
2. Scalability: HPA and Karpenter-driven autoscaling, queue depth metrics, and warm baseline replicas.
3. Security: IRSA, least privilege IAM, network segmentation, private endpoints, and WAF shielding.
4. Observability: Prometheus/Grafana, Fluent Bit logs, OpenTelemetry tracing, health probes, and auto-recovery.
5. Operations: GitHub Actions, Argo CD, Terraform drift checks, runbooks, and team task assignment.